

Underwater Vision-Based Surveillance Using Machine Learning and Artificial Intelligence

Individual Internship Project

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Problem Statement & Objectives

Problem Statement

Underwater images often suffer from:

- Colour distortion
- Haze and scattering
- Low contrast
- Poor visibility
- Illumination variation

Research Question

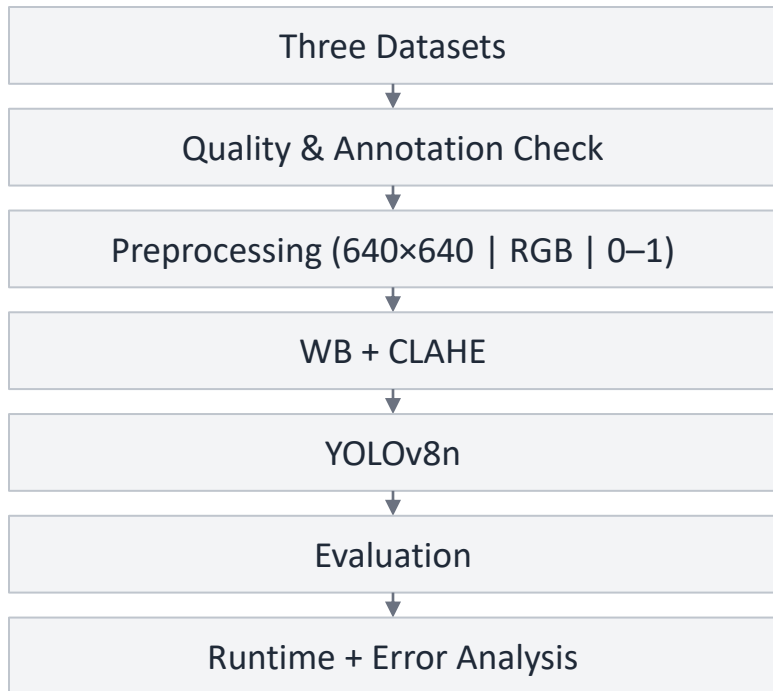
How does underwater image enhancement affect the performance of deep-learning-based object detection across the three supplied underwater datasets?

Objectives

- Analyze all three datasets
- Validate images and YOLO annotations
- Build a preprocessing pipeline
- Implement White Balance + CLAHE
- Train YOLOv8n for detection
- Compare original vs enhanced images
- Evaluate detection performance
- Analyze failures and limitations

Datasets & Methodology

Pipeline



Datasets

Dataset	Classes	Final Configuration
DIATAquarium	11	Original + YOLOv8n
Aquatic Plant	1	WB+CLAHE + YOLOv8n
Well	4	WB+CLAHE + YOLOv8n

Note: DIAT's final configuration used the original image, while WB+CLAHE was used for Aquatic Plant and Well.

Preprocessing, Enhancement & YOLOv8n

Preprocessing

- 640×640 resolution
- RGB format
- Pixel normalization: 0–1
- Raw dataset preserved
- Augmentation only during training

Detection

YOLOv8n outputs the class, bounding box and confidence.



Enhancement

Original → White Balance → CLAHE → Enhanced

White Balance: reduces underwater colour distortion.

CLAHE: improves local contrast while limiting excessive contrast amplification.



Experimental Design & Final Models

Main Experiment

Baseline: Original → YOLOv8n

Enhanced: Original → WB+CLAHE → YOLOv8n

Final Configurations

Dataset	Configuration	LR	Batch	Epochs
DIAT	Original + YOLOv8n	—	16	30
Aquatic Plant	WB+CLAHE + YOLOv8n (H03)	0.0005	32	30
Well	WB+CLAHE + YOLOv8n (H04)	0.001	16	30

Optimizer for H03 / H04: AdamW

Evaluation

Precision, Recall, mAP@0.5, mAP@0.5:0.95, Inference time, Model size

Final Results

Validation Results

Dataset	Precision	Recall	mAP50	mAP50-95
DIAT	90.02%	92.09%	92.73%	65.92%
Aquatic Plant	95.4%	96.1%	98.5%	71.5%
Well	35.7%	40.1%	36.8%	16.8%

Final Test Results

Dataset	Precision	Recall	mAP50	mAP50-95
Aquatic Plant	97.99%	90.26%	96.75%	71.12%
Well	43.00%	37.46%	39.34%	16.15%

Important Finding

WB+CLAHE did not consistently improve detection across datasets. Aquatic Plant: mixed test-set change. Well: lower than baseline on all four test metrics. DIAT: no separate local test set.

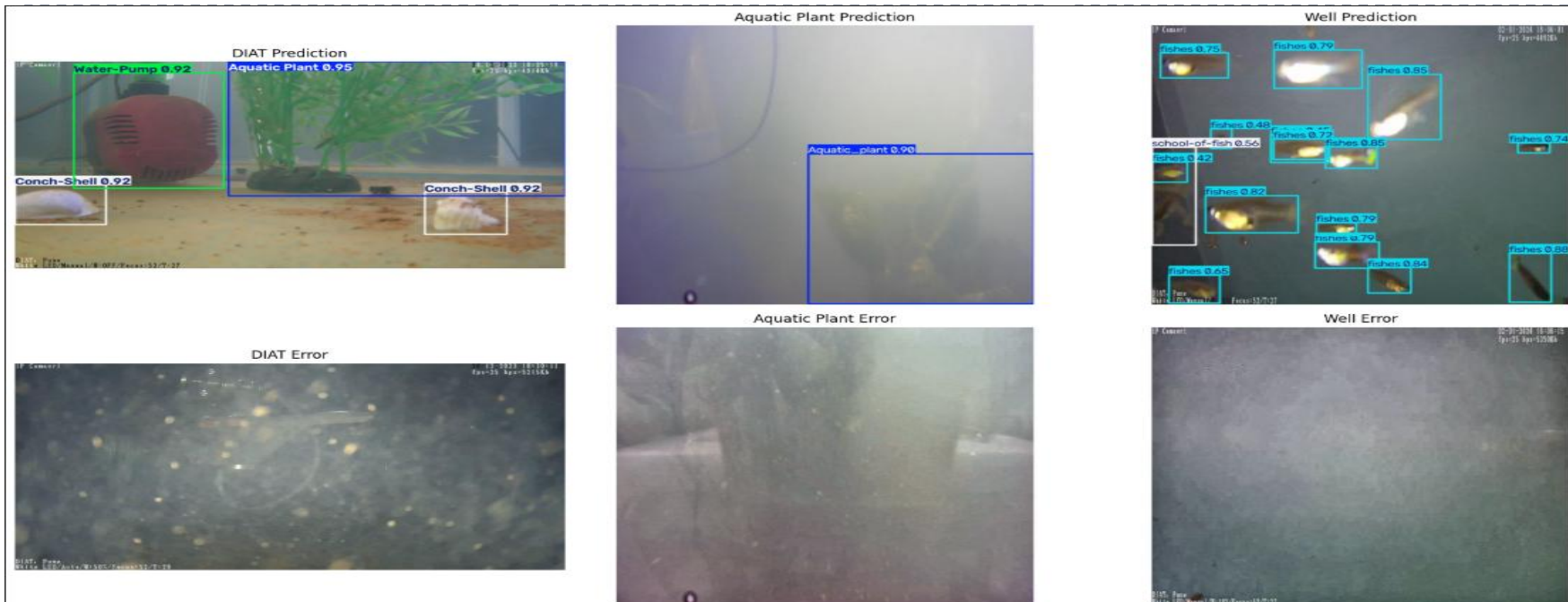
Prediction & Error Analysis

Top: representative validation predictions. Bottom: difficult / error cases.

DIAT

Aquatic Plant

Well



Fish/Fishes confusion; possible missed Camera detection

Missed or low-confidence plant detections

Small/edge objects, stone detection difficulty, possible false positives

Common condition: haze and low underwater visibility.

Limitations, Future Scope & Conclusion

Limitations

- DIAT local copy had no separate test set
- Source overlap was identified
- Well has strong class imbalance
- Error analysis covered representative cases only
- Water-Net / FUnIE-GAN reviewed, not implemented
- Tracking excluded (no video available)

Future Scope

- Evaluate learning-based enhancement methods
- Address class imbalance and rare classes
- Increase training-data diversity
- Use video data for object tracking

Conclusion

White Balance + CLAHE did not consistently improve YOLOv8n detection; its effect was dataset- and class-dependent. The project delivers an end-to-end workflow: Preprocessing → Enhancement → Detection → Evaluation → Error Analysis.